

TraceFlow 1.0.0: Validation Pack

November 13, 2025

Version under test	
Engineer	
Tester	
Test Date	
Initial Report Generation Date	November 13, 2025
Result	
Approved Usage	

Manual tests carried out by:

Signature

Name

Job Title

Date

Validation pack result approved by:

Signature

Name

Job Title

Date

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1 Requirements

These are some general notes on this set of requirements. Other content within subsections are for specific requirements.

1.1 Traceability Matrix

<i>Req ID</i> \ <i>Test ID</i>	TEST-001	TEST-002	TEST-003
REQ-001	✓		
REQ-002	✓		
REQ-003		✓	✓
REQ-004			

1.2 REQ-001: User authentication

Test ID: TEST-001

Users must be able to authenticate using their email address and a password.

1.2.1 Example LaTeX equation

The strength of the user's password should follow the entropy equation:

$$H = L \times \log_2(N)$$

Where H is the entropy, L is the password length, and N is the number of possible symbols.

1.2.2 Example inline code formatting

The ID of the user is, e.g., 1af345e6.

1.3 REQ-002: MRI dataset import

Test ID: TEST-001

The platform must support importing MRI datasets in DICOM format.

1.3.1 Example image (.png)



Figure 1: VoxelFlow Logo

1.4 REQ-003: Image analysis pipeline

Test ID: TEST-002 **Test ID:** TEST-003

- The platform should provide a Python API for building image analysis pipelines.
- The pipelines must be able to process MRI datasets in a compliant way.



Figure 2:

1.4.1 Example flow chart (using mermaid)

1.4.2 Example code block

```
def hello_world():  
    print("Hello_world!")
```

1.5 REQ-004: Output data

The platform must be able to export the results of image analysis pipelines in a standard format, including the following parameters:

2 Manual Acceptance Tests

These are some general notes on this set of test. Other content within subsections are for specific tests.

Tester	
Test Date	
Result	
Observations	

2.1 TEST-001: User authentication

Requirement ID: REQ-001 Requirement ID: REQ-002

2.1.1 Test Steps:

- Navigate to the login page.
- Enter a valid email address and password.
- Click the "Login" button.

2.1.2 Expected Result:

The user is logged in and redirected to the main dashboard.

2.1.3 Test Outcome:

Pass	Fail	Skip
Comments		

2.2 TEST-002: Automatic Playwright test

Requirement ID: REQ-003

2.2.1 Test Steps:

- The autoplaywright block runs the Playwright sample located in the playwright/ directory and captures the video, stdout/stderr, and a 3×3 key frame grid from the test execution.

2.2.2 Expected Result:

The test should pass successfully.

2.2.3 Test Outcome:

Pass ☒	Fail ☐	Skip ☐
Playwright Test: 0001-test-counter		

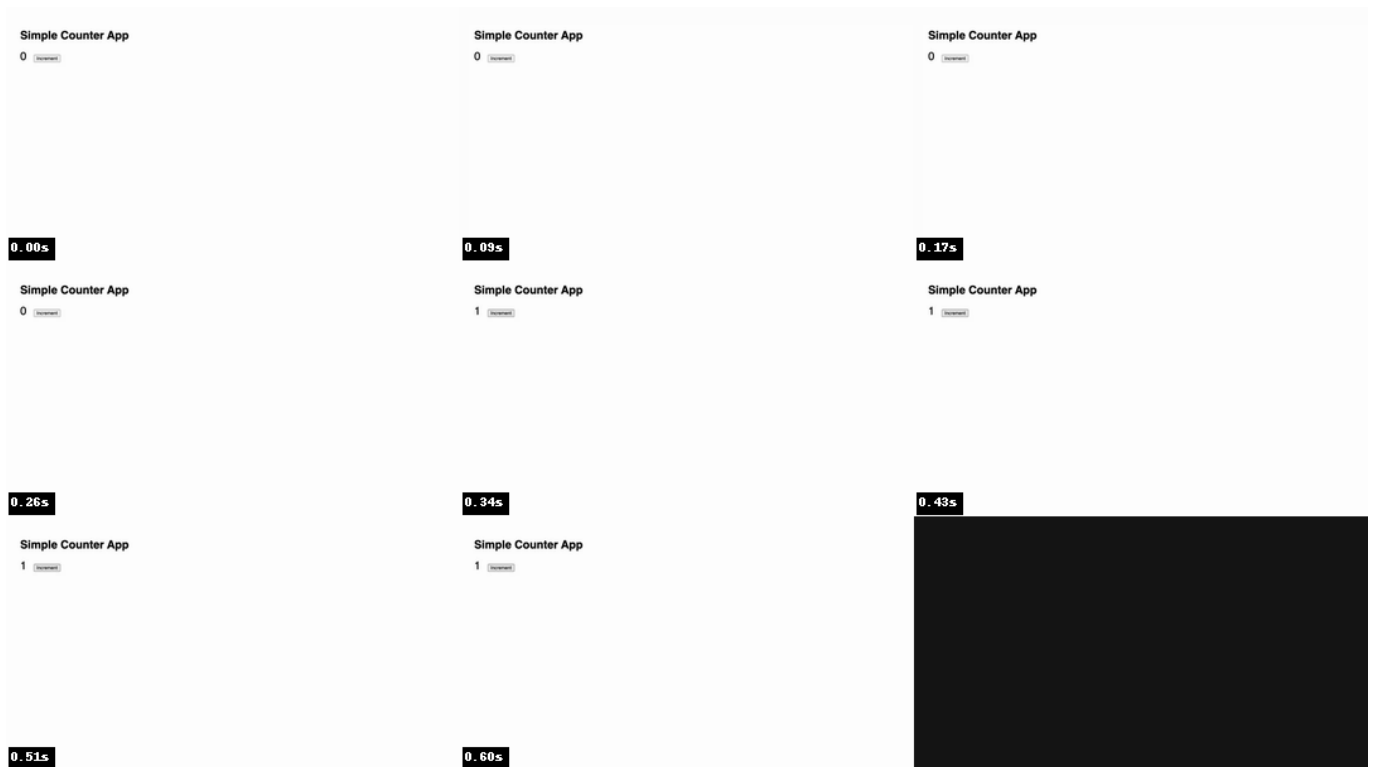


Figure 3: Key frames captured from 0001-test-counter

Running 1 test using 1 worker

```
[PASS] 1 counter.spec.js:5:1 > 0001-test-counter (809ms)
1 passed (1.3s)
```

2.3 TEST-003: Automatic test

Requirement ID: REQ-003

2.3.1 Test Steps:

- This autotest block demonstrates running a simple pytest command as part of the report.

2.3.2 Expected Result:

The test should pass successfully.

2.3.3 Test Outcome:

Pass ☒ Fail ☐ Skip ☐

Playwright Test: 0001-test-counter

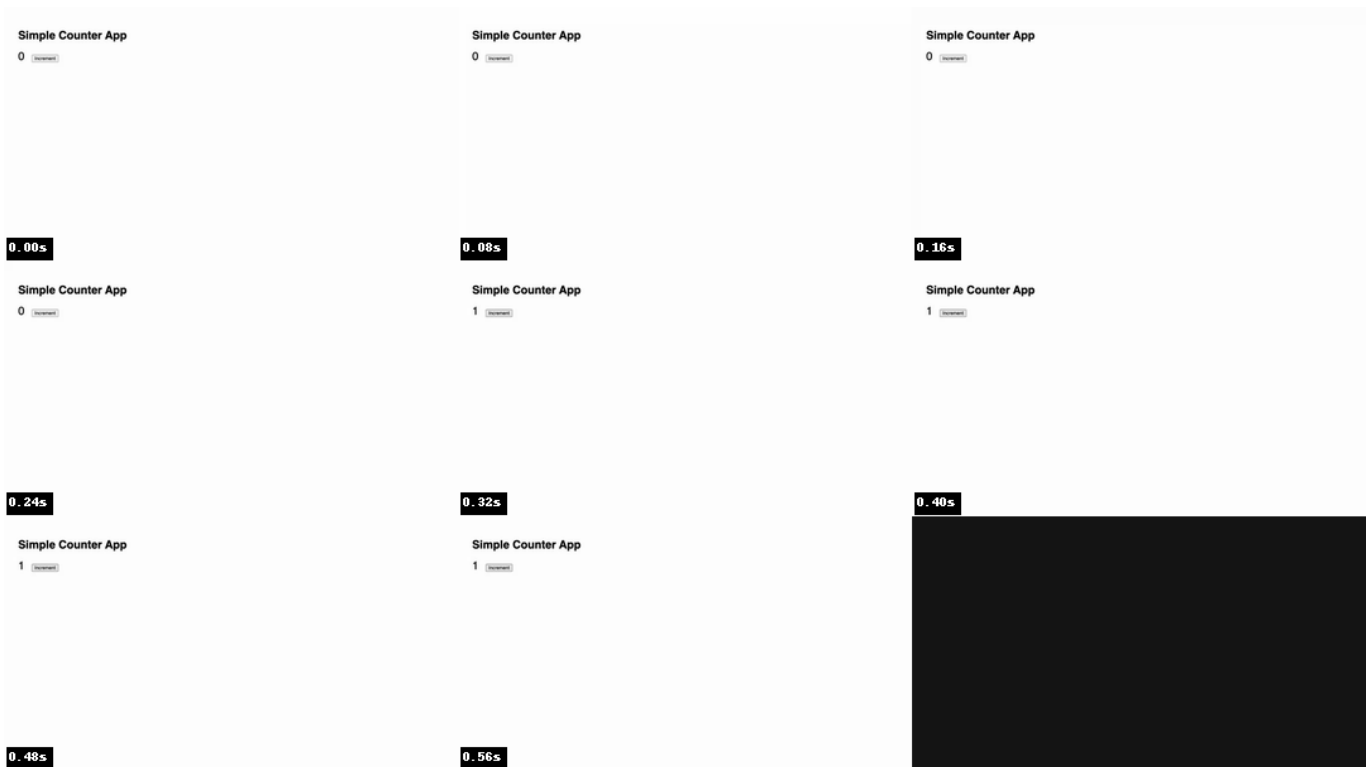


Figure 4: Key frames captured from 0001-test-counter

Running 1 test using 1 worker

```
[PASS] 1 counter.spec.js:5:1 > 0001-test-counter (808ms)
```

1 passed (1.3s)

Pass ☒

Fail ☐

Skip ☐

```
===== test session starts =====
platform darwin -- Python 3.11.13, pytest-9.0.1, pluggy-1.6.0 --
/Users/matthew.heaton/Work/SoftwareDevelopment/TraceFlow/.venv/bin/python3
cachedir: .pytest_cache
rootdir: /Users/matthew.heaton/Work/SoftwareDevelopment/TraceFlow
configfile: pyproject.toml
collecting ... collected 1 item

tests/test_hello_world.py::TestHelloWorld::test_zero Hello world!
PASSED

===== 1 passed in 0.00s =====
```